

Nonlinear optics

11

Practically everything we have been describing so far in this book falls into the realm of linear optics, where it is assumed that properties such as the refractive index, absorption coefficient, and reflectivity are independent of the optical power. This approximation is only valid at low power levels. With a high-power laser, it is possible to enter a different realm of behaviour called **nonlinear optics**. In this subject we consider the consequences of allowing the electric susceptibility, and all the properties that follow from it, to vary with the strength of the electric field of the light beam.

Nonlinear optics is a subject in its own right that has grown in importance as the applications of lasers have become more common. A text such as this would be incomplete without some mention of the types of phenomena that can occur in solid-state materials. The objective here is to give a brief introduction to the subject, based predominantly on the classical dipole oscillator model developed in Chapter 2. It is hoped that this may form a basis for further reading in more comprehensive treatments. A partial list of introductory nonlinear optics texts may be found in the Bibliography.

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11.1 The nonlinear susceptibility tensor

The optical properties of materials are described through the real and imaginary parts of the dielectric constant ϵ_r . The dielectric constant is derived from the polarization $\mathbf{P}$ of the medium according to:

$$\begin{aligned}\mathbf{D} &= \epsilon_0 \mathbf{E} + \mathbf{P}, \\ &= \epsilon_0 \epsilon_r \mathbf{E}.\end{aligned}\tag{11.1}$$

In linear optics, we assume that $\mathbf{P}$ depends linearly on the electric field $\mathbf{E}$ of the light wave, so that we can write:

$$\mathbf{P} = \epsilon_0 \chi \mathbf{E},\tag{11.2}$$

where χ is the electric susceptibility. By combining eqns 11.1 and 11.2 we derive the usual relationship between ϵ_r and χ , namely:

$$\epsilon_r = 1 + \chi.\tag{11.3}$$

In nonlinear optics we consider the possibility that the relationship between $\mathbf{P}$ and $\mathbf{E}$ is more general than that given in eqn 11.2. We start

by considering a nonlinear medium in which the polarization is parallel to the electric field, so that we do not need to consider the vector nature of $\mathbf{P}$ and $\mathbf{E}$ at this stage. We split the polarization P into the first-order linear response $P^{(1)}$, plus a whole series of nonlinear terms of increasing order according to:

$$P^{\text{nonlinear}} = P^{(1)} + P^{(2)} + P^{(3)} + \dots, \quad (11.4)$$

where $P^{(n)}$ is the n th-order polarization.

In analogy with eqn 11.2, we now introduce the **nonlinear susceptibility** $\chi^{\text{nonlinear}}$, and the n th-order nonlinear susceptibility, $\chi^{(n)}$. These are defined by the following equations:

$$\begin{aligned} P^{\text{nonlinear}} &= \epsilon_0 \chi^{\text{nonlinear}} \mathcal{E} \\ &= \epsilon_0 \chi^{(1)} \mathcal{E} + \epsilon_0 \chi^{(2)} \mathcal{E}^2 + \epsilon_0 \chi^{(3)} \mathcal{E}^3 + \dots, \end{aligned} \quad (11.5)$$

where $\mathcal{E}$ is the magnitude of the applied field.

The various terms in eqns 11.4 and 11.5 correspond directly with each other so that

$$P^{(1)} = \epsilon_0 \chi^{(1)} \mathcal{E}, \quad (11.6)$$

$$P^{(2)} = \epsilon_0 \chi^{(2)} \mathcal{E}^2, \quad (11.7)$$

$$P^{(3)} = \epsilon_0 \chi^{(3)} \mathcal{E}^3. \quad (11.8)$$

⋮

By comparing eqns 11.3 and 11.5, we see that:

$$\begin{aligned} \epsilon_r^{\text{nonlinear}} &= 1 + \chi^{\text{nonlinear}}, \\ &= 1 + \chi^{(1)} + \chi^{(2)} \mathcal{E} + \chi^{(3)} \mathcal{E}^2 + \dots, \end{aligned} \quad (11.9)$$

where $\chi^{(1)}$ is just the normal linear susceptibility. Equation 11.9 implies that the dielectric constant depends on the electric field through the nonlinear susceptibilities. Since the optical power is proportional to $\mathcal{E}^2$, this means that ϵ_r also depends on the optical power. Hence properties like the refractive index and absorption coefficient become power-dependent in nonlinear materials.

The different-order nonlinear susceptibilities give rise to a whole host of nonlinear effects. The majority of these phenomena can be attributed to either the $\chi^{(2)}$ or $\chi^{(3)}$ terms in the polarization. These are either called **second-order** or **third-order nonlinear** effects as appropriate. Some of these will be discussed in Sections 11.3 and 11.4.

The well-defined axes of crystalline materials make it necessary to consider that the nonlinear response of the medium may depend on the directions in which the fields are applied. For example, we could apply two optical fields in different directions and then generate a nonlinear polarization along a third direction. This type of behaviour can be described by generalizing eqns 11.7 and 11.8 to allow for the anisotropic

response of the medium. For example, the components of the second-order nonlinear polarization $\mathbf{P}^{(2)}$ can be written in the following form:

$$P_i^{(2)} = \epsilon_0 \sum_{j,k} \chi_{ijk}^{(2)} \mathcal{E}_j \mathcal{E}_k. \quad (11.10)$$

The quantity $\chi_{ijk}^{(2)}$ that appears here is the **second-order nonlinear susceptibility tensor**, and the subscripts i, j , and k correspond to the Cartesian coordinate axes x, y , and z . It will usually be convenient to define these axes so that they coincide with the principal axes of the crystal whenever this is possible.

Equation 11.10 shows that there are nine different contributions for each component of $\mathbf{P}^{(2)}$. For example, the term with $\chi_{xyz}^{(2)}$ gives the nonlinear polarization generated along the x axis when one optical field is applied along the y axis and another along the z axis. We can also generate a nonlinear polarization along the x axis by applying two fields along x via the $\chi_{xxx}^{(2)}$ term, and so on for all nine possible permutations of j and k . At first sight it might therefore appear that we have to measure 27 different quantities in order to quantify the second-order nonlinear response of an anisotropic medium completely. Fortunately, this is not usually the case, because the high degree of symmetry found in crystals requires that many of the terms are zero, and many of the others are the same. This point is developed further in Section 11.3.2.

The third-order nonlinear response of an anisotropic medium is also described by a tensor relationship. We can generalize eqn 11.8 by writing the components of the third-order nonlinear polarization as follows:

$$P_i^{(3)} = \epsilon_0 \sum_{j,k,l} \chi_{ijkl}^{(3)} \mathcal{E}_j \mathcal{E}_k \mathcal{E}_l, \quad (11.11)$$

where $\{i, j, k, l\} \in \{x, y, z\}$, and $\chi_{ijkl}^{(3)}$ is the third-order nonlinear susceptibility tensor. $\chi_{ijkl}^{(3)}$ is a fourth-rank tensor with 81 components. As with the second-order nonlinear susceptibility, symmetry may require that many of these terms are the same or zero.

Example 11.1

Potassium dihydrogen phosphate (KDP) is a uniaxial crystal with a four-fold axis of rotation about the z axis. The tetragonal $\bar{4}2m$ symmetry class of the crystal demands that the only non-zero components of the second-order nonlinear susceptibility tensor are the ones with i, j , and k all different, namely $\chi_{xyz}^{(2)}$, $\chi_{yxz}^{(2)}$, $\chi_{xzy}^{(2)}$, $\chi_{zxy}^{(2)}$, $\chi_{yzx}^{(2)}$, and $\chi_{zyx}^{(2)}$. Furthermore, symmetry also requires that

$$\chi_{xyz}^{(2)} = \chi_{yxz}^{(2)} = \chi_{xzy}^{(2)} = \chi_{yzx}^{(2)}, \quad (11.12)$$

and

$$\chi_{zxy}^{(2)} = \chi_{zyx}^{(2)}. \quad (11.13)$$

It is not necessary that $\mathcal{E}_k$ and $\mathcal{E}_l$ in eqn 11.10 should be derived from different light beams. In many cases there will only be a single beam incident on the crystal, and $\mathcal{E}_k$ and $\mathcal{E}_l$ will just be the components of the electric field resolved along the appropriate axes.

Determine the direction of the nonlinear polarization when a powerful laser beam is propagating along the optic axis.

Solution

If the laser is propagating in the z direction, then the electric field of the light will be polarized along the x or y directions. The nonlinear polarization is therefore given by eqn 11.10 with $\mathcal{E}_z = 0$. This gives:

$$P_i^{(2)} = \epsilon_0 \left(\chi_{ixx}^{(2)} \mathcal{E}_x \mathcal{E}_x + \chi_{ixy}^{(2)} \mathcal{E}_x \mathcal{E}_y + \chi_{iyx}^{(2)} \mathcal{E}_y \mathcal{E}_x + \chi_{iyy}^{(2)} \mathcal{E}_y \mathcal{E}_y \right).$$

If $i = x$ or $i = y$, then all the terms on the right-hand side are zero because $\chi_{ijk}^{(2)}$ is zero unless i, j and k are all different. This means that the nonlinear polarization vector is given by:

$$\begin{aligned} P_x^{(2)} &= 0 \\ P_y^{(2)} &= 0 \\ P_z^{(2)} &= \epsilon_0 \left(\chi_{zzx}^{(2)} \mathcal{E}_x \mathcal{E}_y + \chi_{zzy}^{(2)} \mathcal{E}_y \mathcal{E}_x \right). \end{aligned}$$

We therefore conclude that the nonlinear polarization is pointing along the optic axis, irrespective of the direction of the polarization of the input laser beam.

11.2 The physical origin of optical nonlinearities

The discussion in the previous section gave no indication as to why a particular material should be nonlinear or not. The magnitude of the electric field that binds an electron to an atom is typically around 10^{10} – 10^{11} V m⁻¹. (See Exercise 11.1.) It might therefore be expected that nonlinear effects would become important when the electric field of the light is comparable to this value. From the relationship between the intensity of a light beam and its electric field given by eqn A.44 in Appendix A, namely:

$$I = \frac{1}{2} c \epsilon_0 n \mathcal{E}^2, \quad (11.14)$$

we see that we need optical intensities around 10^{19} W m⁻² to produce fields of this magnitude. Intensities as high as this can just about be achieved with very powerful lasers, but in fact the nonlinear effects set in at much lower intensity levels. This is because we can produce a sizeable macroscopic result by adding together the very small nonlinear effects in a very large number of atoms. This only works if the nonlinear phenomena in all the individual atoms are in phase with each other. This effect is called ‘phase matching’, and is discussed in Section 11.3.3.

The approach taken to explaining the microscopic origin of optical nonlinearities depends on whether the frequency is close to one of the

natural transition frequencies of the atoms or not. If it is, then we are dealing with a **resonant** nonlinear effect, while if it is not, we are considering a **non-resonant** nonlinearity. These two situations are discussed separately below, starting with the non-resonant nonlinearities. It turns out that the non-resonant effects can be explained in terms of the classical oscillator model by introducing anharmonic terms. On the other hand, we need to use a quantum model to account properly for resonant effects.

11.2.1 Non-resonant nonlinearities

In Chapter 2 we explained how we can calculate the response of a medium to electromagnetic waves by assuming that it consists of a series of oscillators with characteristic resonant frequencies. In the near-infrared, visible, or ultraviolet spectral ranges we are normally considering the response due to the electrons. We have been assuming that these are bound to the atoms by harmonic restoring forces such that the displacement induced by the driving field of the light wave is linear. As with most oscillatory systems, this will only be true for small displacements. If the system is driven hard by the strong field of an intense laser beam, the displacements will be large, and it may no longer be valid to assume that the displacement varies linearly with the driving field.

We can account for the non-resonant nonlinear effects by assuming that the electron is bound in an **anharmonic** potential well of the form:

$$U(x) = \frac{1}{2}m_0\omega_0^2x^2 + \frac{1}{3}m_0C_3x^3 + \frac{1}{4}m_0C_4x^4 + \cdots, \quad (11.15)$$

where ω_0 is the natural resonant frequency and $x = 0$ corresponds to the equilibrium position of the electron. It is assumed that $\omega_0^2 \gg C_3x \gg C_4x^2 \dots$, so that it makes sense to carry out the power series expansion, and that the harmonic term dominates for small displacements. The power series expansion is a simplification of the more complicated functional forms that would appear in a real atom. (See Exercise 10.16 for a worked example for the vibrational potential energy.)

We concentrate here on the second-order effects, and consider only the x^3 term in eqn 11.15. The restoring force for displacements from the equilibrium position is given by:

$$F(x) = -\frac{dU}{dx} = -(m_0\omega_0^2x + m_0C_3x^2). \quad (11.16)$$

This shows that the strength of the restoring force now depends on the direction of the displacement: the electron experiences a stronger force for positive displacements than for negative displacements. If we drive the electron with the AC electric field of a light wave, the displacements will be smaller during the positive part of the cycle than for the negative part. Since the dipole moment per unit volume of the medium is equal to $-Nex$, the polarization will likewise be asymmetric in the field direction. Hence the relationship between P and $\mathcal{E}$ will not be linear and will involve powers of $\mathcal{E}$ greater than one.

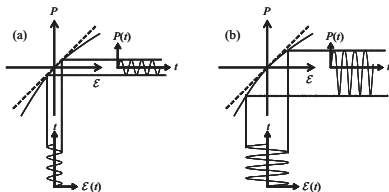


Fig. 11.1 Response of an anharmonic medium to a sinusoidal driving field. The linear dependence between P and $\mathcal{E}$ is shown by the dashed line, while the solid line corresponds to a nonlinear dependence. (a) Small electric field: the departure from linear response is small. (b) Large applied field: the polarization is asymmetric, with larger displacements for negative $\mathcal{E}$.

The nonlinear relationship between P and $\mathcal{E}$ is sketched in Fig. 11.1. For small fields the departure from the linear response is negligible. Hence the polarization closely follows the applied field, as shown in Fig. 11.1(a). However, if the magnitude of the applied field is increased, the response becomes asymmetric, with larger displacements for negative fields. This point is illustrated in Fig. 11.1(b), which shows how the application of a sinusoidal electric field gives a distorted output if the material is nonlinear. It is well known from electrical circuit theory that the distorted output can be described by including higher harmonics. In the case shown in Fig. 11.1(b), the output contains a second harmonic wave with 20% of the amplitude of the fundamental.

This simple discussion shows that the inclusion of the anharmonic term generates a signal at twice the frequency of the applied wave. We can see from eqn 11.7 that this is equivalent to a second-order nonlinearity, because if $\mathcal{E}(t) = \mathcal{E}_0 \sin \omega t$, then

$$\begin{aligned} P^{(2)}(t) &= \epsilon_0 \chi^{(2)} \mathcal{E}_0^2 \sin^2 \omega t \\ &= \frac{1}{2} \epsilon_0 \chi^{(2)} \mathcal{E}_0^2 (1 - \cos 2\omega t). \end{aligned} \quad (11.17)$$

Thus if $\chi^{(2)}$ is non-zero, the medium generates a wave at 2ω when driven at frequency ω . This is the same conclusion as that derived from the consideration of the anharmonic term in the potential, and shows that the two treatments are equivalent.

The relationship between C_3 and $\chi^{(2)}$ can be made more precise by finding an approximate solution to the equation of motion of the electron when driven by an AC electric field at frequency ω . To do this, we proceed as in Section 2.2, but now include the anharmonic term in the restoring force. The equation of motion is thus:

$$m_0 \frac{d^2 x}{dt^2} + m_0 \gamma \frac{dx}{dt} + m_0 \omega_0^2 x + m_0 C_3 x^2 = -e\mathcal{E}, \quad (11.18)$$

where γ accounts for damping. $\mathcal{E}$ is the driving field of the electromag-

netic wave, which is assumed to have the following time dependence:

$$\mathcal{E}(t) = \mathcal{E}_0 \cos \omega t = \frac{1}{2} \mathcal{E}_0 (e^{i\omega t} + e^{-i\omega t}) . \quad (11.19)$$

We have seen above that the inclusion of the C_3 term leads to a response at frequency 2ω in addition to the one at ω . Therefore we write the time dependence of the electron displacement as:

$$x(t) = \frac{1}{2} (X_1 e^{-i\omega t} + X_2 e^{-2i\omega t} + \text{c.c.}) , \quad (11.20)$$

where ‘c.c.’ stands for complex conjugate. We assume that the nonlinear term is small, so that $X_1 \gg X_2$.

On substituting eqn 11.20 into eqn 11.18, we obtain:

$$\begin{aligned} & (-\omega^2 - i\omega\gamma + \omega_0^2)(X_1 e^{-i\omega t} + \text{c.c.})/2 \\ & + (-4\omega^2 - 2i\omega\gamma + \omega_0^2)(X_2 e^{-2i\omega t} + \text{c.c.})/2 \\ & + C_3(X_1^2 e^{-2i\omega t} + 2X_1^* X_2 e^{-i\omega t} + \dots + \text{c.c.})/4 \\ & = \frac{-e\mathcal{E}_0}{2m_0} (e^{-i\omega t} + \text{c.c.}) , \end{aligned} \quad (11.21)$$

where it is assumed that the anharmonic term is small and the ellipsis represents the higher-order cross terms at frequencies other than ω and 2ω . For eqn 11.21 to hold at all times, the coefficients of $e^{\pm i\omega t}$ and $e^{\pm 2i\omega t}$ must be the same on both sides of the equation. We are assuming that the nonlinear response is small, and so we can neglect the term at frequency ω generated from the anharmonic part of the potential. Hence we obtain:

$$X_1 = \frac{-e\mathcal{E}_0}{m_0} \frac{1}{(\omega_0^2 - \omega^2) - i\gamma\omega} . \quad (11.22)$$

This is exactly the same result as eqn 2.9 in Section 2.2, which is hardly surprising, since it represents the linear response of the system. The polarization at frequency ω follows directly:

$$\begin{aligned} P(\omega, t) &= -Nex(\omega, t) \\ &= -Ne(X_1 e^{-i\omega t} + \text{c.c.})/2 \\ &= \epsilon_0 \chi(\omega) \mathcal{E}(t) , \end{aligned} \quad (11.23)$$

where the third line is just the standard definition of the linear susceptibility as in eqn 11.2. By combining eqns 11.19, 11.22, and 11.23, we obtain the usual result for the linear susceptibility:

$$\chi(\omega) = \frac{Ne^2}{m_0 \epsilon_0 [(\omega_0^2 - \omega^2) - i\gamma\omega]} . \quad (11.24)$$

We now solve for X_2 to find the nonlinear response by equating the coefficients of $e^{-2i\omega t}$ in eqn 11.21. This gives:

$$(-4\omega^2 - 2i\omega\gamma + \omega_0^2) \frac{X_2}{2} + \frac{C_3}{4} X_1^2 = 0 , \quad (11.25)$$

It is important to keep track of all the conjugate terms in this model, for otherwise we can lose some of the important cross terms. We focus on the negative frequency terms in order to keep consistency with Section 2.2.

from which we obtain, using eqns 11.22 and 11.24:

$$\begin{aligned} X_2 &= -\frac{C_3 X_1^2}{2(\omega_0^2 - 4\omega^2 - 2i\omega\gamma)} \\ &= -\frac{C_3 e^2 \mathcal{E}_0^2}{2m_0^2(\omega_0^2 - \omega^2 - i\gamma\omega)^2(\omega_0^2 - 4\omega^2 - 2i\omega\gamma)} \\ &= -\frac{m_0 C_3 e_0^3 \chi(\omega)^2 \chi(2\omega)}{2N^3 e^4} \mathcal{E}_0^2. \end{aligned} \quad (11.26)$$

The polarization at frequency 2ω is given by

$$P(2\omega, t) = -Nex(2\omega, t) = -Ne(X_2 e^{-2i\omega t} + \text{c.c.})/2. \quad (11.27)$$

Now in the case we are considering where the polarization at frequency 2ω is generated by nonlinear conversion of the driving field at frequency ω , $P(2\omega)$ will also be given by eqns 11.7 and 11.19 as:

$$P(2\omega, t) = \epsilon_0 \chi^{(2)} \mathcal{E}(t)^2 = \epsilon_0 \chi^{(2)} \left(\frac{\mathcal{E}_0}{2} \right)^2 (e^{-2i\omega t} + \text{c.c.}). \quad (11.28)$$

Thus by combining eqns 11.26–11.28 we obtain the final result (Miller's rule):

$$\chi^{(2)} = \frac{m_0 C_3 \chi(\omega)^2 \chi(2\omega) \epsilon_0^2}{N^2 e^3}. \quad (11.29)$$

We have restricted our attention here to the second-order nonlinearity, but it is obvious that the derivation can be generalized by including higher-order anharmonic terms, which would then explain the origin of higher-order nonlinearities.

This shows that the second-order nonlinear susceptibility is directly proportional to C_3 , the anharmonic term in the equation of motion. Equation 11.29 is reasonably successful in predicting the dispersion of $\chi^{(2)}$ in a large number of crystals. This is because it is found empirically that the anharmonic constant C_3 does not vary very much from material to material.

Equation 11.29 tells us that $\chi^{(2)}$ increases as ω approaches ω_0 , through the frequency dependence of $\chi(\omega)$ given by eqn 11.24. This effect is known as resonance enhancement. In a lightly damped system, the classical treatment breaks down as we get closer to the resonant frequency due to the divergence in $\chi(\omega)$. Hence we have to adopt a different approach if we are close to resonance. This is discussed in the next subsection.

11.2.2 Resonant nonlinearities

The off-resonant nonlinear effects discussed in the previous section are all 'virtual' processes. This means that no real transitions take place because the photon energy does not coincide with any of the transition frequencies of the atoms. The situation is obviously completely different if the laser frequency is in resonance with an atomic transition. In this case, the atoms can absorb photons and make transitions to excited states as the beam propagates through the medium.

The absorption rate is normally determined by the matrix element for the transition and the density of states according to Fermi's golden rule (eqn B.14). This allows us to determine the absorption coefficient for a

particular material at a particular frequency. All this presupposes that the intensity of the light beam on the sample is small. If the intensity is high, the absorption coefficient becomes intensity dependent. Since the absorption coefficient is related to the dielectric constant, this means that the dielectric constant is intensity dependent. In other words, we are dealing with an optical nonlinearity.

The intensity dependence of the absorption rate can be understood through the Einstein B coefficients discussed in Section B.1 of Appendix B. Consider the propagation of an intense laser beam of frequency ν through an absorbing medium. Figure 11.2 illustrates the simplest case to discuss, namely a medium containing atoms with just two levels: level 1 at energy E_1 and level 2 at energy E_2 , where $E_2 > E_1$. We assume that there are N_1 atoms per unit volume in the lower level, and N_2 per unit volume in level 2. The total number of atoms per unit volume is N_0 , where $N_0 = N_1 + N_2$.

We consider the case in which the laser is resonant with the atomic transition frequency of the atoms, such that $h\nu = E_2 - E_1$. The laser beam will be absorbed as it propagates through the medium by transitions upwards from level 1 to level 2. At the same time, photons will be added to the beam by stimulated emission. The stimulated emission rate is not normally considered when discussing the propagation through an absorbing medium because it is assumed that N_2 is negligible. However, if the laser intensity is large, we can no longer make this assumption, because the absorption transitions will create a substantial population in the upper level. This gives rise to a significant stimulated emission rate, and effectively reduces the absorption coefficient.

The reduction of the absorption coefficient due to stimulated emission can be modelled by considering an incremental beam slice of thickness dz as illustrated in Fig. 11.3. The number of photons absorbed per unit time in the incremental slice is given by:

$$\delta N_{\text{absorbed}} = B_{12} N_1 u_\nu g(\nu) \times Adz, \quad (11.30)$$

where u_ν is the energy density of the beam at position z , $g(\nu)$ is the spectral lineshape function of the transition, and A is the beam area. Adz is thus the volume of the slice. The extra factor of $g(\nu)$ in eqn 11.30 compared to eqn B.5 arises from the difference between the beam energy density u_ν (units: J m^{-3}) and spectral energy density $u(\nu)$ (units: $\text{J m}^{-3} \text{Hz}^{-1}$). The Einstein coefficients are traditionally defined when considering the interaction of a broadband light source (e.g. black-body radiation) with a narrow atomic absorption line. Here, by contrast, we are considering a laser beam with a spectral linewidth that is smaller than that of the transition. The inclusion of the spectral lineshape function is required for dimensional consistency and ensures that the transition rate is proportional to the shape of the absorption line.

The stimulated emission rate can be treated in the same way. We see from eqn B.6 that the number of photons added per unit time to the beam is given by:

$$\delta N_{\text{stimulated}} = B_{21} N_2 u_\nu g(\nu) \times Adz. \quad (11.31)$$

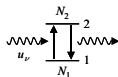


Fig. 11.2 Transitions induced by a resonant laser beam of energy density u_ν . Photons are removed from the beam by absorption transitions from level 1 $\rightarrow$ 2 and are added by stimulated emission transitions from level 2 $\rightarrow$ 1.

For a Lorentzian line, $g(\nu)$ takes the form:

$$g(\nu) = \frac{\Delta\nu/2\pi}{(\nu - \nu_0)^2 + (\Delta\nu/2)^2},$$

where ν_0 is the transition frequency and $\Delta\nu$ its half width. Note that this is normalized such that:

$$\int_0^\infty g(\nu) d\nu = 1.$$

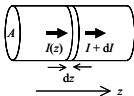


Fig. 11.3 Propagation of a laser beam of area A through an absorbing medium.

Hence the total reduction in the photon number per unit time is given by:

$$\begin{aligned}\delta N_{\text{total}} &= \delta N_{\text{absorbed}} - \delta N_{\text{stimulated}} \\ &= (B_{12}N_1 - B_{21}N_2)u_\nu g(\nu)Adz.\end{aligned}\quad (11.32)$$

This is effectively the net absorption rate.

The intensity of the beam decreases as it propagates through the medium due to absorption. If we take the intensity at position z to be $I(z)$, and the change in the intensity in the incremental slice to be dI , we can write

$$\begin{aligned}AdI &= -\delta N_{\text{total}} \times h\nu, \\ &= -(B_{12}N_1 - B_{21}N_2)u_\nu g(\nu)h\nu Adz.\end{aligned}\quad (11.33)$$

The left-hand side is the beam energy change per unit time in the incremental slice at z . The right-hand side is the change in the photon number per unit time given by eqn 11.32 multiplied by the energy of each photon. Conservation of energy requires that these two quantities must be the same.

Equation 11.33 can be simplified by noting from eqn A.43 in Appendix A that $I = cu_\nu/n$, where n is the refractive index of the medium. Hence we obtain:

$$\frac{dI}{dz} = -\frac{B_{12}(N_1 - N_2)g(\nu)h\nu n}{c}I, \quad (11.34)$$

where we have assumed that the degeneracies of the two levels are the same so that $B_{12} = B_{21}$ (cf. eqn B.10). Equation 11.34 can be compared to the standard definition of the absorption coefficient α given in eqn 1.4, which implies that

$$\frac{dI}{dz} = -\alpha I. \quad (11.35)$$

Hence by comparing eqns 11.34 and 11.35 we see that

$$\alpha = B_{12}(N_1 - N_2)g(\nu)h\nu n/c. \quad (11.36)$$

This shows that the absorption coefficient is proportional to the population difference between the lower and upper levels.

At low intensities we can assume that $N_1 \approx N_0$ and $N_2 \approx 0$. Equation 11.36 then just reduces to the usual result where the absorption coefficient is proportional to the number of atoms in the system. However, at high intensities, the laser pumps a large number of atoms into the upper level so that N_2 increases and N_1 decreases. Hence the absorption coefficient begins to decrease as the population difference between the upper and lower levels decreases.

The decrease of the absorption with the laser power can be characterized by introducing the **saturation intensity** I_s . The absorption coefficient is found experimentally to depend on the intensity I according to the following relationship:

$$\alpha(I) = \frac{\alpha_0}{1 + I/I_s}, \quad (11.37)$$

Note that eqn 11.37 only applies to homogeneously broadened systems, and that the saturation intensity measured for a particular absorption line will depend on the detailed rate constants for that specific transition.

where α_0 is the absorption measured in the linear regime when $I \ll I_s$. A medium which shows the behaviour indicated by eqn 11.37 is called a **saturable absorber**.

At low intensity levels, eqn 11.37 can be expanded to obtain:

$$\alpha(I) = \alpha_0 - (\alpha_0/I_s)I. \quad (11.38)$$

This shows that the absorption decreases linearly with I . Now α is proportional to the imaginary part of ϵ_r (cf. eqns 1.19 and 1.24), and I is proportional to $\mathcal{E}^2$. Hence ϵ_r varies in proportion to $\mathcal{E}^2$, and from eqn 11.9 we see that this is equivalent to a $\chi^{(3)}$ process. In other words, the resonant nonlinearities due to saturable absorption are third-order nonlinear effects. (See Exercise 11.11.)

The analysis of the saturable absorber above applies primarily to the discrete absorption lines found in atomic systems. However, in solid-state materials we are often more interested in the saturation of an absorption band rather than a discrete line. For example, in Section 11.4.7 we present data for the saturable absorption of interband transitions and also of excitons.

In treating the nonlinear saturation of interband absorption, it is useful to adopt a slightly different approach which is based on the Pauli exclusion principle. The dependence of α on $(N_1 - N_2)$ in eqn 11.36 can be considered as a consequence of the Fermi-Dirac statistics of the electrons. For absorption to be possible, the lower level must contain an electron, while the upper level must be empty. Hence the absorption coefficient will obey

$$\alpha = \alpha_0(f_1 - f_2), \quad (11.39)$$

where f_1 and f_2 are the Fermi occupancies of the lower and upper levels respectively. α_0 is the low power absorption when the lower level is full and the upper level empty: that is, when $f_1 = 1$ and $f_2 = 0$. The absorption at high powers is calculated by working out the filling of the levels after a large number of electrons and holes have been excited by the absorption of a laser pulse.

11.3 Second-order nonlinearities

In this section we discuss a few of the more important effects that are associated with the second-order nonlinear susceptibility $\chi^{(2)}$. We begin by considering the general principles of nonlinear frequency mixing, and the effects of the crystal symmetry on the nonlinear coefficients. We then introduce the concept of phase matching, which is crucially important for obtaining large nonlinear signals. Finally, we give a brief discussion of the linear electro-optic effect.

11.3.1 Nonlinear frequency mixing

The second-order nonlinear polarization is given by eqn 11.7. If the medium is excited by sinusoidal waves at frequencies ω_1 and ω_2 with

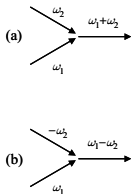


Fig. 11.4 Feynman diagrams for second-order nonlinear frequency mixing processes.

amplitudes $\mathcal{E}_1$ and $\mathcal{E}_2$ respectively, then the nonlinear polarization will be equal to:

$$\begin{aligned} P^{(2)}(t) &= \epsilon_0 \chi^{(2)} \times \mathcal{E}_1 \cos \omega_1 t \times \mathcal{E}_2 \cos \omega_2 t \\ &= \epsilon_0 \chi^{(2)} \mathcal{E}_1 \mathcal{E}_2 [\cos(\omega_1 + \omega_2)t + \cos(\omega_1 - \omega_2)t]/2. \end{aligned} \quad (11.40)$$

This shows that the second-order nonlinear response generates polarization waves at the sum and difference frequencies of the input fields according to:

$$\omega_{\text{sum}} = \omega_1 + \omega_2, \quad (11.41)$$

$$\omega_{\text{diff}} = \omega_1 - \omega_2. \quad (11.42)$$

The medium then re-radiates at ω_{sum} and ω_{diff} , thereby emitting light at frequencies $(\omega_1 + \omega_2)$ and $(\omega_1 - \omega_2)$. This effect is called **nonlinear frequency mixing**. If the frequencies are the same, the sum frequency is twice the input frequency. This effect is called **frequency doubling** or **second-harmonic generation**, and has already been introduced in the discussion of eqn 11.17.

Nonlinear frequency mixing processes can be represented by Feynman diagrams as indicated in Fig. 11.4. Figure 11.4(a) shows the process for sum frequency mixing, while Fig. 11.4(b) represents difference frequency mixing. Conservation of energy applies at each vertex. The negative input frequency at ω_2 for the difference frequency mixing process in Fig. 11.4(b) reflects the fact that $\cos \omega t = (e^{+i\omega t} + e^{-i\omega t})/2$, so that we can represent real waves either with positive or negative frequencies on a Feynman diagram. In quantum-mechanical terms, we would say that the sum frequency mixing process annihilates two input photons at frequencies ω_1 and ω_2 , with the creation of a new photon at frequency ω_{sum} , while difference frequency mixing annihilates one photon at frequency ω_1 and creates two photons, one at frequency ω_2 and the other at ω_{diff} . The creation of the photon at frequency ω_2 in the latter case is stimulated by the presence of a large number of existing photons at frequency ω_2 from the input field.

One of the most important uses for nonlinear optical processes is to generate new frequencies from fixed-wavelength lasers. The most common technique is frequency doubling. In this case, we have a single input beam, and the sum frequency mixing works by taking two photons from the input beam and generating a new photon at the doubled frequency.

Figure 11.5 shows a schematic experimental arrangement for generating the second, third, and fourth harmonics of a Nd:YAG laser operating at 1064 nm. The second harmonic at $1064/2 = 532$ nm is generated by frequency doubling of the fundamental. The second harmonic beam can then be doubled again with an additional nonlinear crystal to generate the fourth harmonic at 266 nm. The third harmonic can be generated by carrying out sum frequency mixing of the fundamental and the second harmonic beams in another nonlinear crystal. These techniques are standard procedures in modern laser physics.

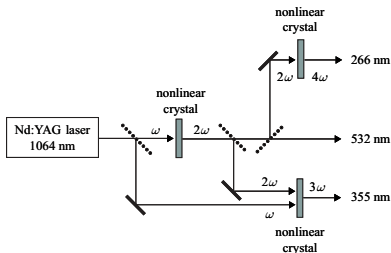


Fig. 11.5 Nonlinear frequency conversion of a Nd:YAG laser operating at 1064 nm. The beam is first doubled to 532 nm. This beam can then either be used as the output, or doubled again to 266 nm. Alternatively, the output at 532 nm can be mixed with the fundamental at 1064 nm to generate the third harmonic at 355 nm. The residual pump beams transmitted through the nonlinear crystals are separated from the harmonics by filters that are not shown in the diagram.

When the two input fields are at the same frequency, we see from eqn 11.42 that the difference frequency is zero. This effect is called **optical rectification**, and refers to the phenomenon by which a static electric field is produced from fields of optical frequencies. The **Pockels effect**, which is also known as the **linear electro-optic effect**, is the reverse of this process, and is discussed further in Section 11.3.4.

The sum frequency mixing process shown in Fig. 11.4(a) can work the other way round as well. In this case, we bring in a single input field at frequency ω and create two new photons at frequencies of ω_1 and ω_2 , where $\omega_1 + \omega_2 = \omega$. This process is called **down conversion**. It is apparent that the output frequencies generated by down-conversion are not uniquely defined. Any combination of frequencies that satisfies the conservation of energy requirements can in principle be generated. However, the number of photons emitted at any particular frequency will only be large if the phase matching conditions discussed in Section 11.3.3 are satisfied.

Down conversion can be used to amplify a weak beam by a process called **parametric amplification**. If we introduce a weak 'signal' field at frequency ω_s in the presence of a strong pump field at frequency ω , it can generate an 'idler' field at $\omega_i = \omega - \omega_s$ by difference frequency mixing with the pump field. These new idler photons then generate more signal photons by further mixing with the pump field. The process then repeats itself. If the phase matching conditions are satisfied, it is possible to transfer power from the pump beam to the signal and idler beams. Furthermore, if the crystal is inside an optical cavity which is resonant with either ω_s or ω_i , then oscillation can occur. This process is called **parametric oscillation**, and can lead to the generation of intense beams at tunable frequencies even though we started from a fixed-wavelength laser.

A very interesting aspect of down-conversion processes is that the photons are always created in pairs. This means that the photon statistics at frequency ω_1 are directly correlated with those at ω_2 . This gives rise to a whole host of quantum optical effects. For example, the correlated photon pair can be used as the basis of a 'heralded' single-photon source: the detection of a photon in one beam 'heralds' the presence of a single photon in the other.

Table 11.1 Second-order nonlinear effects. The third column lists the frequencies of the light beams incident on the nonlinear crystal, while the fourth gives the frequency of the output beam or the nonlinear polarization. A frequency of zero indicates a DC electric field.

Effect	Alternative name	Input frequencies	Output frequencies
Frequency doubling	Second harmonic generation	ω	2ω
Optical rectification		ω	0
Down conversion	Linear electro-optic effect	ω	ω_1, ω_2
Sum frequency mixing		ω_1, ω_2	$(\omega_1 + \omega_2)$
Difference frequency mixing		ω_1, ω_2	$ \omega_1 - \omega_2 $
Pockels effect		$\omega, 0$	ω

The different types of second-order nonlinear effects considered in this section are summarized in Table 11.1.

11.3.2 Effect of crystal symmetry

The second-order nonlinear susceptibility introduced in eqn 11.10 is a third rank tensor with 27 components. Fortunately, it is not necessary to measure all of these components to determine the nonlinear response of the medium. It is immediately obvious that some of the 27 components are the same. For example the term $\chi_{xyz}^{(2)}\mathcal{E}_y\mathcal{E}_z$ must be the same as $\chi_{xzy}^{(2)}\mathcal{E}_z\mathcal{E}_y$ because the response of the medium cannot depend on the mathematical ordering of the fields. Hence there are in fact only 18 physically distinct components of the nonlinear susceptibility. This means that we can write the nonlinear response in a simpler form in terms of a contracted tensor called the **nonlinear optical coefficient tensor** d_{ij} . Written out explicitly, the components of the nonlinear polarizations are given by:

$$\begin{pmatrix} P_x^{(2)} \\ P_y^{(2)} \\ P_z^{(2)} \end{pmatrix} = \begin{pmatrix} d_{11} & d_{12} & d_{13} & d_{14} & d_{15} & d_{16} \\ d_{21} & d_{22} & d_{23} & d_{24} & d_{25} & d_{26} \\ d_{31} & d_{32} & d_{33} & d_{34} & d_{35} & d_{36} \end{pmatrix} \begin{pmatrix} \mathcal{E}_x\mathcal{E}_x \\ \mathcal{E}_y\mathcal{E}_y \\ \mathcal{E}_z\mathcal{E}_z \\ 2\mathcal{E}_y\mathcal{E}_z \\ 2\mathcal{E}_x\mathcal{E}_x \\ 2\mathcal{E}_x\mathcal{E}_y \end{pmatrix}. \tag{11.43}$$

By comparing this with eqn 11.10 we see that $d_{11} = \epsilon_0\chi_{xxx}^{(2)}$, $d_{14} = \epsilon_0\chi_{xyz}^{(2)}$, etc.

In many crystals the nonlinear optical coefficient tensor can be further simplified because the crystal symmetry requires that many of the terms are zero, and many others are the same. This is a consequence of Neumann's principle which states that the macroscopic physical properties of a crystal must be invariant under the symmetry operations of the crystal. (See Section 1.5.1.) This makes the task of characterizing the nonlinear material much easier than it might seem at first.

The simplest case to consider is that of a centrosymmetric crystal. This is a crystal that has inversion symmetry. Suppose we generate a

nonlinear polarization in such a crystal by using a single applied field, $\mathcal{E}$. The components of $\mathbf{P}^{(2)}$ will be given by eqn 11.10. If we now reverse the direction of the electric field, nothing happens because:

$$\begin{aligned} P_i^{(2)}(-\mathcal{E}) &= \epsilon_0 \sum_{j,k} \chi_{ijk}^{(2)}(-\mathcal{E}_j)(-\mathcal{E}_k) \\ &= \epsilon_0 \sum_{j,k} \chi_{ijk}^{(2)} \mathcal{E}_j \mathcal{E}_k \\ &= P_i^{(2)}(+\mathcal{E}). \end{aligned} \quad (11.44)$$

However, since the crystal has inversion symmetry, we know from Neumann's principle that we must get the same physical result by keeping the field in the original direction and inverting the crystal. In terms of the coordinate axes of the inverted crystal, all the components of $\mathcal{E}$ and $\mathbf{P}^{(2)}$ change sign. Therefore, for the inverted crystal we have:

$$-P_i^{(2)} = \epsilon_0 \sum_{j,k} \chi_{ijk}^{(2)}(-\mathcal{E}_j)(-\mathcal{E}_k). \quad (11.45)$$

The only way that eqn 11.45 can be compatible with eqn 11.44 is if $\chi_{ijk}^{(2)} = 0$ for all permutations of i, j, k . Hence we conclude that the second-order nonlinear susceptibility of centrosymmetric crystals is zero, so that $d_{ij} = 0$ for all i and j .

If the crystal does not possess inversion symmetry, then some of the components of d_{ij} will be non-zero. In triclinic crystals with the lowest possible symmetry, it will be necessary to specify all 18 values of d_{ij} to describe the nonlinear response fully. At the other extreme, in materials with the zinc-blende structure (class $\bar{4}3m$), it is only necessary to specify one value because the very high degree of crystal symmetry requires that the only non-zero terms are d_{14} , d_{25} , and d_{36} , and that these three are all identical.

In crystals with intermediate symmetry, it will be necessary to specify a varying number of physically different terms in d_{ij} . For example, we already considered the uniaxial nonlinear crystal KDP (potassium dihydrogen phosphate, KH_2PO_4) in Example 11.1. KDP belongs to the tetragonal crystal class $\bar{4}2m$ and has four-fold rotational symmetry about the z axis. The non-zero components of $\chi_{ijk}^{(2)}$ specified in eqns 11.12 and 11.13 imply that the only non-zero components of d_{ij} are d_{14} , d_{25} , and d_{36} , with d_{14} equal to d_{25} . Hence we can characterize the nonlinear response completely by making just two separate measurements to determine d_{14} and d_{36} . In other types of crystal, there will be different relationships between the coefficients of d_{ij} . Tables of these relationships are given in books on crystallography and nonlinear optics, and representative values of the nonlinear coefficients of several important crystals are given in Table 11.2. Exercise 11.6 works through a specific example for the case of an orthorhombic nonlinear crystal.

A more obvious but less rigorous argument why $\chi^{(2)} = 0$ in centrosymmetric crystals follows from eqn 11.15. A centrosymmetric crystal must have $U(x) = U(-x)$ since the physical properties are invariant on inverting the crystal. This means that $C_3 = 0$, and hence, from eqn 11.29, that $\chi^{(2)} = 0$.

The non-zero terms in d_{ij} for the cubic crystal are derived from the terms of the type $\chi_{xyz}^{(2)}$. It is obvious that $\chi_{xyz}^{(2)} = \chi_{yxx}^{(2)} = \chi_{xxy}^{(2)}$ in a cubic crystal in which the x , y , and z axes are equivalent.

Table 11.2 Nonlinear coefficients of a number of important nonlinear crystals. The nonlinear coefficients are all measured at 1064 nm. Data from Tang (1995) and Klein et al. (2003).

Crystal	Symmetry	Transmission range (nm)	Nonlinear coefficient (pm/V)
KDP (KH_2PO_4)	$\bar{4}2\text{m}$	200–1500	$d_{36} = 0.39$ $d_{14} = 0.4$
KTP (KTiOPO_4)	mm2	350–4400	$d_{31} = 6.5$ $d_{32} = 5.0$ $d_{33} = 14$ $d_{24} = 7.6$ $d_{15} = 6.1$
BBO ($\beta\text{-BaB}_2\text{O}_4$)	3m	190–2500	$d_{22} = 2.1$ $d_{31} = 0.26$
LBO (LiB_3O_5)	mm2	160–2600	$d_{32} = 1.2$
LiNbO ₃	3m	400–5000	$d_{31} = -4.8$ $d_{33} = -30$ $d_{22} = 2.3$

11.3.3 Phase matching

Nonlinear effects are generally small, and we therefore need a long length of the nonlinear medium to obtain a useful nonlinear conversion efficiency. For this to work, we need that the phases of the nonlinear waves generated throughout the whole crystal are all the same so that the fields add together coherently. When this is achieved, we are in a regime called **phase matching**. As we shall see below, phase matching does not normally occur, and can only be achieved if the nonlinear crystal is orientated in a very precise direction.

We can see why phase matching is an important issue by considering a simple example. Suppose we wish to use a nonlinear crystal to double the frequency of a Nd:YAG laser from 1064 nm to 532 nm, as shown schematically in Fig. 11.5. All materials are dispersive to some extent, and this means that the refractive index at 532 nm will be different to that at 1064 nm. Therefore, the second harmonic waves at 532 nm will propagate with a different phase velocity to the fundamental at 1064 nm. This means that the second harmonic waves generated at the front will arrive at the back of the crystal at a different time to the fundamental, and so the 532 nm waves generated at the back of the crystal will be out of phase with those from the front.

The phase mismatch introduced by the frequency doubling process can be calculated from the wave vectors of the two waves. If the beams are travelling in the z direction, then the nonlinear waves will propagate

as $\exp(ik^{(2\omega)}z)$, where $k^{(2\omega)}$ is the wave vector at frequency 2ω . On the other hand, the fundamental beam propagates as $\exp(ik^{(\omega)}z)$, where $k^{(\omega)}$ is the wave vector at frequency ω . Now since $P^{(2)} \propto \mathcal{E}^2$, the nonlinear polarization at a given point in the medium will be created with a phase of $[\exp(ik^{(\omega)}z)]^2 = \exp(i2k^{(\omega)}z)$. Hence the phase difference $\Delta\Phi$ between the nonlinear waves created at a distance z into the crystal and those created at the front of the crystal is given by

$$\Delta\Phi = (k^{(2\omega)} - 2k^{(\omega)})z. \quad (11.46)$$

We introduce the coherence length l_c for the nonlinear process as the distance over which the phase mismatch becomes equal to 2π :

$$(k^{(2\omega)} - 2k^{(\omega)}) \times l_c = 2\pi. \quad (11.47)$$

This can be rewritten in terms of the refractive indices $n^{2\omega}$ and n^ω at the two frequencies as:

$$\frac{2\omega}{c} [n^{2\omega} - n^\omega] l_c = 2\pi. \quad (11.48)$$

Hence

$$l_c = \frac{\pi c}{\omega[n^{2\omega} - n^\omega]} = \frac{\lambda}{2[n^{2\omega} - n^\omega]}, \quad (11.49)$$

where λ is the vacuum wavelength of the fundamental beam. Taking a typical example with $\lambda = 1 \mu\text{m}$ and $n^{2\omega} - n^\omega \sim 10^{-2}$ we find that $l_c \sim 50 \mu\text{m}$.

Equation 11.49 shows us that only the waves emitted within a very short distance of the surface will add together coherently. This clearly greatly restricts the efficiency of the nonlinear conversion process, because only a very short length of the nonlinear crystal is actually useful. The situation would be completely different if we could somehow arrange that $n^{2\omega} = n^\omega$. In this case the nonlinear waves generated throughout the whole crystal would all have the same phase and would thus add together coherently. This is the phase-matching condition.

At first sight it might seem that there is no way to satisfy the condition $n^{2\omega} = n^\omega$ in any material with normal dispersion properties. However, this neglects the fact that the anisotropic crystals that are used for nonlinear mixing are birefringent. This opens new possibilities to balance the dispersion against the birefringence. For example, consider a uniaxial crystal with normal dispersion and negative birefringence, such that $n^{2\omega} > n^\omega$ and $n_e < n_o$. (See Sections 2.4 and 2.5.1.) In such a crystal it is possible to obtain phase matching by propagating the beam at frequency 2ω as an extraordinary ray, and the beam at ω as an ordinary ray. It is shown in Example 11.2 below that phase matching can then be achieved for a very specific orientation of the crystal.

The phase-matching condition can be given an intuitive physical interpretation if we notice that if $n^{2\omega} = n^\omega$, then $k^{(2\omega)} = 2k^{(\omega)}$. This corresponds to momentum conservation in the nonlinear process. In the more general case when a photon of wave vector $\mathbf{k}$ is generated by mixing

two photons with wave vectors $\mathbf{k}_1$ and $\mathbf{k}_2$, the phase-matching condition can be written as

$$\mathbf{k} = \mathbf{k}_1 + \mathbf{k}_2. \quad (11.50)$$

In down conversion where one photon is split into two output photons, the phase-matching condition of eqn 11.50 applies to each pair of photons created in the process.

Example 11.2

The ordinary and extraordinary refractive indices of a uniaxial crystal are n_o and n_e respectively. A laser beam is propagating at an angle θ to the optic (z) axis as shown in Fig. 2.13. The laser is linearly polarized along the x direction.

The arrangement with the pump beam as an ordinary ray (o-ray) and the second harmonic as an extraordinary ray (e-ray) is called **type I phase matching**. In **type II phase matching**, one of the pump photons is an o-ray and the other is an e-ray, while the second harmonic is again an e-ray.

- (a) Show that there is an angle θ at which the phase-matching condition can be met for second-harmonic waves polarized as extraordinary rays. (b) Evaluate the phase-matching angle for potassium dihydrogen phosphate (KDP) at the wavelength of a Nd:YAG laser (1064 nm). The relevant refractive indices for KDP are: $n_o(1064 \text{ nm}) = 1.494$, $n_o(532 \text{ nm}) = 1.512$, and $n_e(532 \text{ nm}) = 1.471$.

Solution

- (a) The general condition for phase matching is that

$$n^{2\omega} = n^\omega. \quad (11.51)$$

The fundamental wave is polarized along the x axis, and so its refractive index is n_o^ω , irrespective of θ . The refractive index for the second harmonic waves polarized as extraordinary rays is given by the result of Exercise 2.16, namely:

$$\frac{1}{n(\theta)^2} = \frac{\sin^2 \theta}{n_e^2} + \frac{\cos^2 \theta}{n_o^2}, \quad (11.52)$$

where n_o and n_e are evaluated at 2ω . Hence the phase-matching condition given in eqn 11.51 is met when

$$\frac{1}{(n_o^\omega)^2} = \frac{\sin^2 \theta}{(n_e^{2\omega})^2} + \frac{\cos^2 \theta}{(n_o^{2\omega})^2}. \quad (11.53)$$

- (b) On inserting the appropriate refractive indices into eqn 11.53, we have:

$$\frac{1}{1.494^2} = \frac{\sin^2 \theta}{1.471^2} + \frac{\cos^2 \theta}{1.512^2}.$$

This is satisfied with $\theta = 41^\circ$. To achieve phase matching we therefore hold the crystal in a gimbal mount and carefully rotate its orientation until the optic axis is at 41° to the direction of the laser beam.

11.3.4 Electro-optics

In electro-optics, a DC electric field is used to change the refractive index of an optical material. As we saw in Section 2.5.2, the refractive index change can be either linear in the field (the Pockels effect) or quadratic (the Kerr effect). In this subsection we focus on the linear electro-optic effect, leaving relevant discussion of the quadratic effect to Section 11.4.3.

It was pointed out in Section 11.3.1 that the linear electro-optic effect can be considered as a type of second-order nonlinearity in which the frequency of the driving field is equal to zero. This has an immediate consequence. Since the second-order susceptibility is zero if the crystal has inversion symmetry, then the linear electro-optic effect can only be observed in crystals that lack inversion symmetry. In fact, the same materials (e.g. KDP, LiNbO₃) are frequently used in both nonlinear optics and electro-optics.

The general treatment of the linear electro-optic effect starts from the index ellipsoid of the anisotropic medium. The index ellipsoid describes the variation of the refractive index with the direction of the electric field of the light, and takes the form:

$$\frac{x^2}{n_x^2} + \frac{y^2}{n_y^2} + \frac{z^2}{n_z^2} = 1, \quad (11.54)$$

where n_x , n_y , and n_z are the refractive indices measured for light polarized along the principal axes, namely $\hat{x}$, $\hat{y}$, and $\hat{z}$ respectively. The refractive index measured for light polarized along the (x, y, z) direction is then given by:

$$n = \sqrt{x^2 + y^2 + z^2}. \quad (11.55)$$

As mentioned in Section 2.5.2, the basic effect of the electric field is to change the anisotropy of the crystal. This effect can be quantified by re-writing the index ellipsoid in the following form:

$$\left(\frac{1}{n^2}\right)_1 x^2 + \left(\frac{1}{n^2}\right)_2 y^2 + \left(\frac{1}{n^2}\right)_3 z^2 + 2\left(\frac{1}{n^2}\right)_4 yz + 2\left(\frac{1}{n^2}\right)_5 xz + 2\left(\frac{1}{n^2}\right)_6 xy = 1. \quad (11.56)$$

On comparing eqns 11.54 and 11.56, it is immediately apparent that at zero applied field we must have that:

$$\left(\frac{1}{n^2}\right)_1 = \frac{1}{n_x^2}, \quad \left(\frac{1}{n^2}\right)_2 = \frac{1}{n_y^2}, \quad \left(\frac{1}{n^2}\right)_3 = \frac{1}{n_z^2}, \quad (11.57)$$

and

$$\left(\frac{1}{n^2}\right)_4 = \left(\frac{1}{n^2}\right)_5 = \left(\frac{1}{n^2}\right)_6 = 0. \quad (11.58)$$

The changes in the index ellipsoid are determined by the components of the **electro-optic coefficient tensor** r_{ij} , defined by:

$$\Delta\left(\frac{1}{n^2}\right)_i = \sum_{j=1}^3 r_{ij} \mathcal{E}_j. \quad (11.59)$$

See Born & Wolf (1999), Nye (1957) or Yariv (1997) for a discussion of the index ellipsoid (also called the 'indicatrix' in some older texts) and its use in linear optics. In uniaxial crystals with their optic axis along z , we set $n_x = n_y = n_o$ and $n_z = n_e$ in eqn 11.54, but in biaxial crystals, all three indices are different.

Written out explicitly, this becomes:

$$\begin{pmatrix} \Delta(1/n^2)_1 \\ \Delta(1/n^2)_2 \\ \Delta(1/n^2)_3 \\ \Delta(1/n^2)_4 \\ \Delta(1/n^2)_5 \\ \Delta(1/n^2)_6 \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \\ r_{41} & r_{42} & r_{43} \\ r_{51} & r_{52} & r_{53} \\ r_{61} & r_{62} & r_{63} \end{pmatrix} \begin{pmatrix} \mathcal{E}_x \\ \mathcal{E}_y \\ \mathcal{E}_z \end{pmatrix}. \quad (11.60)$$

As with the nonlinear optical coefficient tensor, symmetry requires that many of the tensor components are zero, and that some (or all) of the non-zero components have the same magnitude.

Let us consider the case of a KDP (KH_2PO_4) crystal, which has tetragonal $\bar{4}2m$ symmetry. With no field applied, KDP is a uniaxial birefringent crystal, with ordinary and extraordinary refractive indices of n_o and n_e respectively. (See Section 2.5.1.) It is shown in Example 11.3 below that the application of a DC field along the optic (i.e. z) axis rotates the principal axes of the crystal by 45° , as illustrated in Fig. 11.6(a). The modified refractive indices with respect to the new principal axes ($\hat{x}', \hat{y}', \hat{z}'$) are given by:

$$\begin{aligned} n_1(\mathcal{E}) &= n_o + n_o^3 r_{63} \mathcal{E} / 2, \\ n_2(\mathcal{E}) &= n_o - n_o^3 r_{63} \mathcal{E} / 2, \\ n_3(\mathcal{E}) &= n_e. \end{aligned} \quad (11.61)$$

This shows that the field induces anisotropy in the x - y plane, where there was none before. The effect of the field is therefore to induce birefringence, as mentioned previously in Section 2.5.2.

Let us consider a ray propagating in the z direction incident on a KDP crystal. With no field applied, this is an ordinary ray: it experiences no birefringence, and emerges with its polarization unchanged. However, when the field is applied, the ray now experiences birefringence on account of the induced anisotropy in the x - y plane. This means that the output polarization will be different to the input polarization.

Let us assume that the input beam is linearly polarized along the y axis. The $\mathcal{E}$ -vector can be resolved into equal components along the x' and y' axes, which experience refractive indices of $n_1(\mathcal{E})$ and $n_2(\mathcal{E})$ respectively. The phase difference between them is given by:

$$\Delta\Phi(\mathcal{E}) = \frac{2\pi}{\lambda} |n_1 - n_2| L = \frac{2\pi}{\lambda} n_o^3 r_{63} \mathcal{E} L, \quad (11.62)$$

where L is the length of the crystal, and λ is the vacuum wavelength. In general, the output beam will be elliptically polarized. However, if $\Delta\Phi = \pi/2$, then the electro-optic crystal acts as a quarter wave plate and turns the beam into circularly polarized light. Alternatively, if $\Delta\Phi = \pi$, then the electro-optic crystal acts as a half wave plate, and the beam emerges linearly polarized with its polarization rotated by 90° .

The modification of the polarization induced by the field can be used to make a light amplitude modulator, as shown in Fig. 11.6(b). The

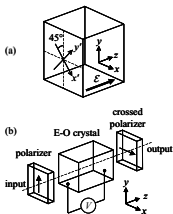


Fig. 11.6 (a) Rotated principal axes x' and y' for a crystal with tetragonal $\bar{4}2m$ symmetry when a field is applied along the z axis. (b) An electro-optic (E-O) modulator based on the crystal shown in part (a). This arrangement is often called a Pockels cell.

crystal is mounted between crossed polarizers, so that there is no output without a field applied. When a beam is incident with its polarization parallel to the axis of the input polarizer and a voltage is applied, the beam emerges from the electro-optic crystal with a different polarization. This means that some of the light now passes through the second polarizer. The output of the device is thus controlled by the voltage applied to the crystal, with the maximum output occurring when the voltage applied induces a π phase shift.

The principles that we have illustrated for the KDP crystal with tetragonal $\bar{4}2m$ symmetry can be applied to other crystals with different symmetry classes. Exercise 11.8 considers the case of trigonal crystals with $3m$ symmetry (e.g. LiNbO_3) and explains how they can be used as electro-optic phase modulators, while Exercise 11.9 considers crystals with cubic symmetry. Table 11.3 lists the values of the electro-optic coefficients for some of the crystals commonly used in modulators.

Table 11.3 Electro-optic coefficients r_{ij} of common electro-optic crystals at 633 nm. KDP and ADP are abbreviations for KH_2PO_4 and $\text{NH}_4\text{H}_2\text{PO}_4$ respectively. After Yariv (1997).

Crystal	Symmetry	r_{ij} (pm/V)
KDP	$\bar{4}2m$	$r_{41} = 8$ $r_{63} = 11$
ADP	$\bar{4}2m$	$r_{41} = 23$ $r_{63} = 7.8$
LiNbO_3	$3m$	$r_{13} = 9.6$ $r_{22} = 6.8$ $r_{33} = 31$ $r_{51} = 33$
KNbO_3	$2mm$	$r_{13} = 28$ $r_{42} = 380$ $r_{51} = 105$
GaP	$\bar{4}3m$	$r_{41} = -0.97$

Example 11.3

Consider a uniaxial crystal of the tetragonal $\bar{4}2m$ class (e.g. KDP) with ordinary and extraordinary refractive indices equal to n_o and n_e respectively. When the optic axis is defined to be along the z direction, Neumann's principle requires that the electro-optic coefficient tensor of crystals with this symmetry class takes the form:

$$\mathbf{r} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ r_{41} & 0 & 0 \\ 0 & r_{41} & 0 \\ 0 & 0 & r_{36} \end{pmatrix}. \quad (11.63)$$

- Write down the modified index ellipsoid when an electric field of magnitude $\mathcal{E}$ is applied along the z axis.
- Show that the principal axes of the crystal with the field applied are rotated by -45° about the z axis compared to the case with no field applied.
- Hence find the modified refractive indices of the crystal measured relative to the rotated principal axes.

Solution

(a) We have a uniaxial crystal with a refractive index of n_o for light polarized along the x or y directions, and n_e for light polarized along z . With no field applied, the index ellipsoid is therefore of the form:

$$\frac{x^2 + y^2}{n_o^2} + \frac{z^2}{n_e^2} = 1.$$

The modified ellipsoid in the presence of the field is found by substituting into eqn 11.60 with $\mathcal{E}_x = \mathcal{E}_y = 0$ and $\mathcal{E}_z = \mathcal{E}$. There is only one non-zero term, namely:

$$\Delta \left(\frac{1}{n^2} \right)_6 = r_{63} \mathcal{E}.$$

The modified index ellipsoid is therefore given by eqn 11.56 as:

$$\frac{x^2}{n_o^2} + \frac{y^2}{n_o^2} + \frac{z^2}{n_e^2} + 2r_{63}\mathcal{E}xy = 1.$$

(b) We define new axes rotated by -45° about the z axis such that:

$$\begin{aligned} x' &= (x - y)/\sqrt{2} \\ y' &= (x + y)/\sqrt{2} \\ z' &= z, \end{aligned}$$

which implies that $x = (x' + y')/\sqrt{2}$, $y = (-x' + y')/\sqrt{2}$ and $z = z'$. On substituting into the modified index ellipsoid we obtain:

$$\frac{x'^2 + y'^2}{n_o^2} + \frac{z'^2}{n_e^2} + r_{63}\mathcal{E}(-x'^2 + y'^2) = 1.$$

This can be written in the form:

$$\frac{x'^2}{n_1^2} + \frac{y'^2}{n_2^2} + \frac{z'^2}{n_3^2} = 1,$$

if we set:

$$\begin{aligned} 1/n_1^2 &= 1/n_o^2 - r_{63}\mathcal{E}, \\ 1/n_2^2 &= 1/n_o^2 + r_{63}\mathcal{E}, \\ 1/n_3^2 &= 1/n_e^2. \end{aligned}$$

This shows that x' and y' are the modified principal axes of the crystal with the field applied.

(c) It is immediately apparent from part (b) that $n_3(\mathcal{E}) = n_e$. We can find $n_1(\mathcal{E})$ and $n_2(\mathcal{E})$ by writing:

$$\begin{aligned} n_1(\mathcal{E}) &= n_o(1 - n_o^2 r_{63} \mathcal{E})^{-1/2}, \\ n_2(\mathcal{E}) &= n_o(1 + n_o^2 r_{63} \mathcal{E})^{-1/2}, \end{aligned}$$

and assuming that the field-induced changes are small compared to n_o . The approximation $(1 + x)^{-1/2} = (1 - x/2)$ is then valid, and so we conclude:

$$\begin{aligned} n_1(\mathcal{E}) &= n_o + n_o^3 r_{63} \mathcal{E} / 2, \\ n_2(\mathcal{E}) &= n_o - n_o^3 r_{63} \mathcal{E} / 2, \\ n_3(\mathcal{E}) &= n_e. \end{aligned}$$

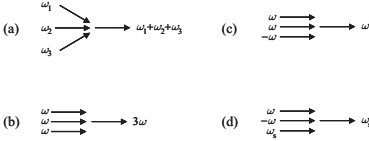


Fig. 11.7 Feynman diagrams for third-order nonlinear processes. (a) Four-wave mixing. (b) Frequency tripling. (c) The optical Kerr effect. (d) The Raman effect.

11.4 Third-order nonlinear effects

Third-order nonlinear effects are particularly important in isotropic media, such as gases, liquids, and glasses. This is because an isotropic medium possesses inversion symmetry, and all the components of $\chi_{ijk}^{(2)}$ must therefore be zero. (See Section 11.3.2.) Hence the lowest-order nonlinear susceptibility with non-vanishing components is $\chi^{(3)}$. In this section we start by giving an overview of third-order nonlinear phenomena, and then focus on isotropic media in more detail, mentioning optical fibres as a specific example. Finally we discuss resonant third-order nonlinear effects in semiconductors.

11.4.1 Overview of third-order phenomena

A third-order nonlinear polarization is generated when three input fields are applied to the nonlinear medium. If the input fields are at frequencies ω_1 , ω_2 , and ω_3 , then the nonlinear polarization is given by eqn 11.8 or more generally by eqn 11.11. In the simplest case without considering the tensor aspect of the susceptibility, this gives:

$$P^{(3)}(t) = \epsilon_0 \chi^{(3)} \times \mathcal{E}_1 \cos \omega_1 t \times \mathcal{E}_2 \cos \omega_2 t \times \mathcal{E}_3 \cos \omega_3 t, \quad (11.64)$$

where $\mathcal{E}_1$, $\mathcal{E}_2$, and $\mathcal{E}_3$ are the amplitudes of the three waves. Hence the frequency ω_4 of the nonlinear polarization must satisfy:

$$\omega_4 = \omega_1 + \omega_2 + \omega_3, \quad (11.65)$$

where the frequencies on the right-hand side can be either positive or negative. This reflects the fact that $\cos \omega t = (e^{+i\omega t} + e^{-i\omega t})/2$ and therefore contains both positive and negative frequency terms.

Figure 11.7 shows the Feynman diagrams for several third-order nonlinear processes. Figure 11.7(a) gives the diagram for the general process, with three input photons corresponding to the driving fields and one output photon corresponding to the nonlinear polarization. The output frequency is the sum of the input frequencies, as required by conservation of photon energy. Since four photons are involved, the general phenomenon is often called **four-wave mixing**. In the discussion below,

As with the second-order processes, the negative frequencies represent the creation of photons in the nonlinear process. This is a legitimate outcome because photons are bosons and hence the nonlinear interaction can stimulate the creation of photons at the input frequency as well as causing annihilation of input photons.

Table 11.4 Third-order nonlinear effects. The third column lists the frequencies of the light beams incident on the nonlinear medium, while the fourth gives the frequency of the output beam or the nonlinear polarization. A frequency of zero indicates a DC electric field. Note that there are several other third-order nonlinear effects such as two-photon absorption which have not been included in this table.

Effect	Alternative names	Input frequencies	Output frequencies
Generic four-wave mixing		$\omega_1, \omega_2, \omega_3$	$ \pm \omega_1 \pm \omega_2 \pm \omega_3 $
Frequency tripling	Third-harmonic generation	ω	3ω
Optical Kerr effect	Degenerate four-wave mixing Nonlinear refraction Self-phase modulation	ω	ω
DC Kerr effect	Quadratic electro-optic effect	$\omega, 0$	ω
Stimulated four-wave mixing	Stimulated Raman scattering Stimulated Brillouin scattering	ω, ω_0	ω_0

we concentrate on the three specific examples of four-wave mixing illustrated in Figs 11.7(b)–(d), namely: frequency tripling, the optical Kerr effect, and stimulated Raman scattering. There are, of course, many other important third-order nonlinear phenomena, but we do not have space to discuss them all here. The main effects that we consider are summarized in Table 11.4.

11.4.2 Frequency tripling

In practice it is usually easier to generate the third harmonic of a laser beam by two second-order processes according to the scheme shown in Fig. 11.5, rather than by using a single third-order conversion using the $\chi^{(3)}$ nonlinearity.

Figure 11.7(b) shows the Feynman diagram for frequency tripling. This is the equivalent of frequency doubling for a $\chi^{(2)}$ process. Three collinear fields at the same frequency are incident on the medium from a single input laser beam. With $\omega_1 = \omega_2 = \omega_3 = +\omega$, we find from eqn 11.65 that the output frequency is equal to 3ω . In other words, the nonlinear process directly generates the third harmonic of the fundamental. As with frequency doubling, the conversion efficiency will only be large if the phase-matching condition determined by momentum conservation is satisfied. (See Section 11.3.3.) Frequency tripling experiments are useful for the spectroscopic information they give about the magnitude of $\chi^{(3)}$ and its relation to the atomic transitions of the medium.

11.4.3 The optical Kerr effect and the nonlinear refractive index

Figure 11.7(c) shows the Feynman diagram for the optical Kerr effect. In this process we have just a single beam at frequency ω incident on the nonlinear medium, and the nonlinear interaction produces a third-order polarization at the same frequency as the input laser beam. This works by using $\omega_1 = \omega_2 = +\omega$ and $\omega_3 = -\omega$ in eqn 11.65. No phase-matching problems occur in this case because the nonlinear polarization is at the same frequency as the driving fields and thus the fields are in phase throughout the whole medium. Since the frequencies of all four photons

are the same, the optical Kerr effect is sometimes called degenerate four-wave mixing.

One of the main consequences of the optical Kerr effect is that the refractive index begins to depend on the intensity of the beam. This can be seen by calculating the change of the dielectric constant produced by the light. From eqn 11.9 we see that the dielectric constant in a nonlinear medium with $\chi^{(2)} = 0$ is given by:

$$\epsilon_r^{\text{nonlinear}} = 1 + \chi^{(1)} + \chi^{(3)}\mathcal{E}^2, \quad (11.66)$$

where $\chi^{(1)}$ is the linear susceptibility and $\mathcal{E}$ is the optical electric field amplitude. We split this into its linear and nonlinear parts by writing:

$$\epsilon_r^{\text{nonlinear}} = \epsilon_r + \Delta\epsilon, \quad (11.67)$$

where

$$\epsilon_r = 1 + \chi^{(1)} \quad (11.68)$$

and

$$\Delta\epsilon = \chi^{(3)}\mathcal{E}^2. \quad (11.69)$$

ϵ_r is the usual relative permittivity for the linear regime and $\Delta\epsilon$ is the change caused by the nonlinear process. In a non-absorbing medium, the refractive index n is equal to the square root of the relative permittivity (cf. eqn A.31). Hence we may write:

$$n = (\epsilon_r + \Delta\epsilon)^{1/2} = \sqrt{\epsilon_r} + \frac{\Delta\epsilon}{2\sqrt{\epsilon_r}} \equiv n_0 + \Delta n, \quad (11.70)$$

where we have assumed that $\Delta\epsilon \ll \epsilon_r$ in the second equality, and in the third we have split the refractive index into its linear part $n_0 = \sqrt{\epsilon_r}$ and its nonlinear part Δn . On comparing eqns 11.69 and 11.70 we find that

$$n = n_0 + \frac{\chi^{(3)}\mathcal{E}^2}{2n_0} = n_0 + \frac{\chi^{(3)}}{n_0^2 c \epsilon_0} I, \quad (11.71)$$

where we have used the proportionality between I and $\mathcal{E}^2$ given by eqn 11.14 in the second identity.

We now introduce the **nonlinear refractive index** n_2 according to:

$$n(I) = n_0 + n_2 I. \quad (11.72)$$

By comparing eqns 11.71 and 11.72 we find that:

$$n_2 = \frac{1}{n_0^2 c \epsilon_0} \chi^{(3)}. \quad (11.73)$$

This shows that n_2 is directly proportional to $\chi^{(3)}$, and hence that third-order nonlinearities cause the refractive index to vary with the intensity.

Table 11.5 lists the measured values of the nonlinear refractive index for a number of materials. It is apparent that the nonlinear refractive index can be either positive or negative. For a given material, n_2 is

Table 11.5 Nonlinear refractive index n_2 of selected materials. E_g : band gap, λ : wavelength of measurement, $\hbar\omega$: photon energy, n_0 : linear refractive index. Data from Sheik-Bahae et al. (1991) and DeSalvo et al. (1996).

Material	E_g (eV)	λ (nm)	$\hbar\omega/E_g$	n_0	n_2 ($\text{m}^2 \text{W}^{-1}$)
GaAs	1.42	1064	0.87	3.47	-3.3×10^{-17}
CdTe	1.44	1064	0.81	2.84	-2.9×10^{-17}
AlGaAs	1.57	850	0.93	3.30	-2.5×10^{-17}
AlGaAs	1.57	830	0.95	3.30	-8.9×10^{-17}
AlGaAs	1.57	810	0.98	3.30	-3.3×10^{-16}
ZnTe	2.26	1064	0.52	2.79	1.3×10^{-17}
CdS	2.42	1064	0.48	2.34	5.0×10^{-18}
CdS	2.42	532	0.96	2.34	-6.1×10^{-17}
ZnSe	2.58	1064	0.45	2.48	2.9×10^{-18}
ZnSe	2.58	532	0.90	2.70	-6.2×10^{-18}
Fused silica (SiO_2)	7.8	1064	0.15	1.48	2.1×10^{-20}
Fused silica (SiO_2)	7.8	532	0.30	1.48	2.2×10^{-20}
Fused silica (SiO_2)	7.8	355	0.45	1.48	2.4×10^{-20}
Fused silica (SiO_2)	7.8	266	0.60	1.50	7.8×10^{-20}

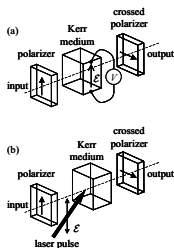


Fig. 11.8 (a) A Kerr cell based on the DC Kerr effect. (b) A Kerr gate based on the optical Kerr effect.

positive at low photon energies, and turns negative for $\hbar\omega \gtrsim 0.7E_g$, where E_g is the band gap. Note that $|n_2|$ is resonantly enhanced as the photon energy approaches E_g , as demonstrated by the data for AlGaAs.

It is apparent from eqn 11.71 that Δn is proportional to $\mathcal{E}^2$ in the optical Kerr effect. This is reminiscent of the DC Kerr effect considered in Section 2.5.2, and explains the name ‘optical Kerr effect’. Both types of Kerr effect are examples of quadratic electro-optic effects. In the DC Kerr effect, $\mathcal{E}$ is an applied DC field, whereas in the optical Kerr effect, $\mathcal{E}$ is the electric field of the light. The DC Kerr effect can in fact be considered as a $\chi^{(3)}$ process in which $\omega_1 = \omega_2 = 0$ and $\omega_3 = \omega$. This contrasts with the *linear* electro-optic effect considered in Section 11.3.4, which is a $\chi^{(2)}$ process.

The analogy between the DC and optical Kerr effects is illustrated in Fig. 11.8. Fig. 11.8(a) shows a conventional Kerr cell based on the DC Kerr effect, which operates in a similar way to the Pockels cell shown in Fig. 11.6(b). The cell comprises a Kerr medium inserted between crossed polarizers. With no external field applied, the Kerr medium is isotropic, and no light is transmitted through the crossed polarizers. When the voltage is applied, birefringence is induced in proportion to $\mathcal{E}^2$ (see eqn 2.51), and the light that emerges from the Kerr medium is no longer vertically polarized. This means that some light is now transmitted through the second polarizer, allowing the cell to be used as an intensity modulator.

Fig. 11.8(b) illustrates a Kerr gate based on the optical Kerr ef-

fect. The principle of operation is the same as the Kerr cell shown in Fig. 11.8(a), except that now there is no applied voltage, and the birefringence is induced by an intense laser pulse via the optical Kerr effect. The gate therefore only transmits light while the laser pulse is impinging on the crystal. By using an ultrafast pulse from a mode-locked laser, it is possible to use this arrangement to produce a very fast (e.g. 1 ps) optical shutter.

11.4.4 Stimulated Raman scattering

Figure 11.7(d) shows the Feynman diagram for the **stimulated Raman effect**. A weak beam at frequency ω_s is incident on the medium together with a powerful pump beam at frequency ω . With $\omega_1 = +\omega$, $\omega_2 = -\omega$, and $\omega_3 = \omega_s$, we see from eqn 11.65 that the nonlinear wave is at frequency ω_s . Hence the presence of a field at ω_s generates more photons at the same frequency through nonlinear mixing with the pump field. The beam at frequency ω_s can then experience gain by the same sort of parametric mixing that was discussed in Section 11.3.1. The process becomes a Raman effect if we tune the frequency of ω_s such that $\omega - \omega_s = \Omega$, where Ω is the frequency of one of the vibrational modes of the medium. The nonlinear susceptibility is resonantly enhanced in these conditions because the two frequencies are strongly coupled together through the natural vibrations of the medium.

Stimulated Raman scattering was discovered very soon after the invention of the laser. In 1962 E.J. Woodbury and W.K. Ng observed that an intense beam was generated at 766 nm when passing a strong beam from a ruby laser at 694.3 nm through a nitrobenzene cell. Analysis of these results showed that the difference in the frequency of the two photons corresponded exactly with one of the vibrational modes of the molecule at 4.0×10^{13} Hz. The same phenomenon has subsequently been observed in many different liquids and gases, and also in solids.

In solids the scattering can be mediated either by the optical or the acoustic vibrational modes. In the former case it is the Raman-active LO and TO phonons at $q = 0$ that are involved, where q is the wave vector of the phonon, as discussed in Section 10.5.2. This gives rise to discrete frequency shifts analogous to those observed in molecules. In the latter case, it is the acoustic phonons that are involved, and the process is usually called **stimulated Brillouin scattering**. The frequency shift caused by stimulated Brillouin scattering depends on the angle through which the light is scattered, as determined by the energy and wave vector conservation laws. (See eqn 10.32.)

11.4.5 Isotropic third-order nonlinear media

The third-order nonlinear response is described by the nonlinear susceptibility tensor $\chi_{ijkl}^{(3)}$ defined in eqn 11.11. This has 81 elements, many of which are the same or zero in materials with a high degree of symmetry. In a completely isotropic medium such as a gas, there are 21 non-zero

The spontaneous Raman scattering processes discussed in Section 10.5 can be related to the stimulated effects we are considering here by making an analogy with spontaneous and stimulated radiative emission. Spontaneous radiative emission can be considered to be a stimulated process that is triggered by a vacuum photon from the zero-point fluctuations of the quantized electromagnetic field. In the same way we can regard spontaneous Raman scattering as a stimulated nonlinear process instigated by vacuum photons. In fact, when we generate stimulated Raman beams by passing a laser through a suitable medium, there is usually no initial field at frequency ω_s to start the process. This field has to come from spontaneous Raman scattering, which itself is instigated by vacuum fluctuations. Hence the generation of a stimulated Raman beam is considered to start from the zero-point fluctuations of the field.

Table 11.6 Non-zero components of the third-order nonlinear susceptibility in an isotropic medium.

$$\begin{aligned}
\chi_{xxxx}^{(3)} &= \chi_{yyyy}^{(3)} = \chi_{zzzz}^{(3)} \\
\chi_{xxyy}^{(3)} &= \chi_{yyxx}^{(3)} = \chi_{xxzz}^{(3)} = \chi_{zzxx}^{(3)} = \chi_{yyzz}^{(3)} = \chi_{zzyy}^{(3)} \\
\chi_{xyxy}^{(3)} &= \chi_{yxyx}^{(3)} = \chi_{xzzx}^{(3)} = \chi_{zxzx}^{(3)} = \chi_{yzyz}^{(3)} = \chi_{zyzy}^{(3)} \\
\chi_{xyyx}^{(3)} &= \chi_{yxxy}^{(3)} = \chi_{xzzz}^{(3)} = \chi_{zzxz}^{(3)} = \chi_{yzzz}^{(3)} = \chi_{zyyz}^{(3)}
\end{aligned}$$

elements that are listed in Table 11.6. The interrelationships listed in the table suggest that there are four independent values, but this is not the case because it can be shown that the susceptibilities must also satisfy:

$$\chi_{xxxx}^{(3)} = \chi_{xxyy}^{(3)} + \chi_{xyxy}^{(3)} + \chi_{xyyx}^{(3)}. \quad (11.74)$$

This means that there are in fact only three independent elements. Furthermore, if we are well below the natural resonance frequencies of the atoms, it will also be true that:

$$\chi_{xxyy}^{(3)} = \chi_{xyxy}^{(3)} = \chi_{xyyx}^{(3)} = \frac{1}{3} \chi_{xxxx}^{(3)}. \quad (11.75)$$

This result is only valid at low frequencies, and is called **Kleinman symmetry**. In this limit, there are thus only two physically different third-order nonlinear susceptibilities.

Glasses are perhaps the most important examples of isotropic optical materials. We usually use glasses at wavelengths where they are transparent. We are therefore well below the band gap in the ultraviolet spectral region, and the Kleinman symmetry condition given in eqn 11.75 will normally hold.

When an intense laser beam propagates through a glass, it can alter the refractive index by the optical Kerr effect in accordance with eqn 11.72. This produces a nonlinear phase shift given by:

$$\Delta\Phi^{\text{nonlinear}} = \frac{2\pi}{\lambda} \Delta n L = \frac{2\pi}{\lambda} n_2 I L, \quad (11.76)$$

where λ is the vacuum wavelength, L is the length of the medium and I is the intensity. The laser beam therefore alters its own phase, an effect called **self-phase modulation**.

11.4.6 Nonlinear propagation in optical fibres and solitons

Self-phase modulation effects can be observed very clearly in optical fibres, even though the nonlinear refractive index is very small. This is

Glasses are isotropic because they do not possess a crystal structure and therefore have no preferred axes. Doped glasses have optical transitions in the visible spectral region (see Section 1.4.5) and the Kleinman symmetry condition does not hold at frequencies close to the band gap of the dopant.



Fig. 11.9 A group of runners on a mattress is an analogy of a soliton pulse in an optical fibre. The indentation of the group slows the faster runners and speeds the slower ones, thereby compensating for the tendency of the group to break up. In the same way, the nonlinear phase shift of an intense short pulse can compensate for the pulse broadening effect due to the dispersion of the optical fibre. After Mollenauer and Gordon (1994), reprinted with permission from Plenum Publishers.

because the beam is focused to a very small area in the fibre, so that the intensity is very high even at moderate power levels. Large phase shifts can then be achieved by using long lengths of fibre.

The subject of **solitons** is a very interesting aspect of the propagation of light pulses down an optical fibre in the nonlinear regime. A short laser pulse must necessarily contain a spread of frequencies in order to satisfy the Fourier transform limit given approximately by eqn 9.6. Since the glass that makes up the fibre is dispersive (see Fig. 2.10), the different frequency components of the pulse will experience slightly different refractive indices. This means that their velocities will be different, and the pulse will gradually broaden in time as it propagates down the fibre. (See Exercise 2.15.) This becomes a serious problem if we are trying to transmit a sequence of closely-spaced data pulses down the fibre. The soliton effect discovered by John Scott Russell in 1834 can eliminate this problem. Russell noticed that the bow wave of a Scottish canal barge did not disperse when the amplitude of the wave was large enough. He successfully explained his observation by realizing that the dispersion of the water wave was being balanced by the nonlinear effects due to the large amplitude. The same phenomenon can occur in optical fibres.

Solitons can be observed in optical fibres at frequencies where the dispersion is negative. This is because the n_2 of the fibre is positive in the near infrared spectral region, and so the dispersion must be negative if the two effects are to cancel. Most glasses have positive dispersion at optical frequencies due to the electronic absorption in the ultraviolet. However, as noted in Section 2.4, the dispersion of the SiO_2 used to make optical fibres is zero at $1.3\ \mu\text{m}$ and becomes negative for longer wavelengths. This can be seen in the data shown in Fig. 2.7. This means that we are in the right regime to observe solitons at $1.55\ \mu\text{m}$, which is the preferred wavelength for telecommunication systems because the losses of the fibre are smallest.

The basic physics of soliton propagation can be understood in a simple way without recourse to heavy mathematics by reference to the cartoon given in Fig. 11.9. This shows a group of athletes crossing a mattress, with the faster runners at the front. The group creates a valley in the mattress that retards the faster runners but helps the slower ones. On a

hard surface the group would break up with the faster runners leaving the slower ones behind, but the valley in the mattress opposes this effect and keeps the group together. In the right conditions the nonlinear phase shift of an intense pulse in an optical fibre can have a similar effect, creating a trapped light pulse similar to the trapped group of runners in the cartoon. The resulting pulse can travel indefinitely long distances along the fibre without significant broadening. This is a very useful property for long-distance optical fibre telecommunication systems.

Example 11.4

A laser pulse at $1.55\text{ }\mu\text{m}$ of intensity 10^{12} W m^{-2} is propagating down an optical fibre of length 100 m . Calculate the nonlinear phase shift if the nonlinear refractive index is $2 \times 10^{-20}\text{ m}^2\text{ W}^{-1}$.

Solution

We calculate the nonlinear phase shift from eqn 11.76 with $\lambda = 1.55 \times 10^{-6}\text{ m}$, $n_2 = 2 \times 10^{-20}\text{ m}^2\text{ W}^{-1}$, and $I = 10^{12}\text{ W m}^{-2}$. This gives:

$$\Delta\Phi^{\text{nonlinear}} = \frac{2\pi}{1.55 \times 10^{-6}} \times (2 \times 10^{-20}) \times 10^{12} \times 100 = 8.1.$$

The nonlinear phase shift is thus 2.6π . This example shows that the nonlinear phase shift can be very large, even though the nonlinearity is small. This occurs because of the very large values of L that can be used in optical fibres.

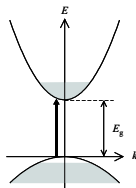


Fig. 11.10 Band-filling nonlinearity in an excited semiconductor. The interband absorption transition shown by the arrow in the diagram is blocked because there are no electrons in the valence band to absorb, and the destination states in the conduction band are filled.

11.4.7 Resonant nonlinearities in semiconductors

In Section 11.2.2 we explained how the absorption coefficient of an absorbing medium is expected to depend on the intensity at high power levels. This phenomenon has been studied extensively in semiconductors, with a particular emphasis on developing nonlinear switching devices for applications in optical information processing.

The simplest mechanism that can cause saturable absorption in a semiconductor is illustrated in Fig. 11.10. This shows the band diagram of a direct gap semiconductor when a large number of electrons have been excited from the conduction band to the valence band. The electrons are excited by absorption of an intense laser pulse with photon energy greater than the band gap. The electrons fill up the states at the bottom of the conduction band, leaving empty states at the top of the valence band. This blocks further interband absorption transitions at photon energies close to the band gap such as the one shown in the figure, because there are no electrons in the valence band to be excited, and furthermore, the destination states in the conduction band are full. Therefore, as we turn up the intensity of the exciting laser, we find that

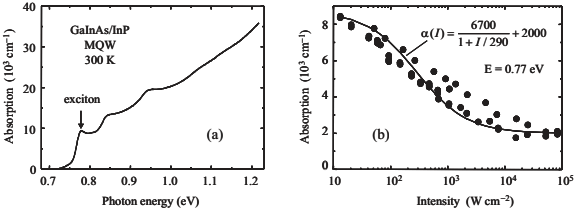


Fig. 11.11 Saturable excitonic absorption in a semiconductor multiple quantum well. The sample contained 30 Ga_{0.53}In_{0.47}As quantum wells of width 15.4 nm with InP barriers. The temperature was 300 K. Part (a) shows the linear absorption of the sample, while (b) shows the intensity dependence of the absorption at 0.77 eV. This photon energy corresponds to the first excitonic peak in the linear absorption spectrum, and is identified by the arrow in (a). The solid line in (b) is a fit to the data using eqn 11.77. After Westland et al. (1987) and Fox et al. (1987), © American Institute of Physics, reprinted with permission.

the absorption gradually saturates as the bands fill up. This effect is therefore called a **band-filling nonlinearity**.

Another mechanism that can cause the absorption to depend on the intensity is the saturation of excitons. This effect was described in Section 4.4. At high carrier densities the Coulomb force holding the excitons together is screened, and the electron and hole states near $k = 0$ required for the formation of excitons are filled. Both of these effects cause bleaching of the exciton absorption. If the carriers are excited by the absorption of a laser beam, the excitonic absorption will decrease with increasing laser intensity.

Figure 11.11 shows the intensity dependence of the exciton absorption in a GaInAs multiple quantum well sample at room temperature. The linear absorption spectrum is given in Fig. 11.11(a). The step-like spectrum expected for the 2-D quantum wells is clearly apparent in the data, together with the broadened peaks due to the excitons. Figure 11.11(b) shows the saturation of the absorption coefficient α at the first exciton peak as a function of intensity I . The intensity dependence of α is given approximately by:

$$\alpha(I) = \alpha_b + \frac{\alpha_0}{1 + I/I_s}, \quad (11.77)$$

with $\alpha_b = 2000 \text{ cm}^{-1}$, $\alpha_0 = 6700 \text{ cm}^{-1}$, and $I_s = 290 \text{ W cm}^{-2}$. The α_b term in eqn 11.77 represents an unsaturable background absorption, while the second term represents a saturable absorption of the form given in eqn 11.37.

The intensity dependence of α shown in Fig. 11.11(b) gives rise to a very large change in the refractive index. (See Exercise 11.15.) The nonlinear refractive index n_2 associated with the exciton saturation is $\sim 10^{-8} \text{ m}^2 \text{ W}^{-1}$. This is many orders of magnitude larger than the values

Quantum wells are good materials to demonstrate excitonic nonlinearities because they show strong excitonic absorption even at room temperature. See Section 6.4.4 for further details.

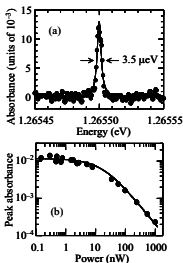


Fig. 11.12 (a) Excitonic absorption line of a single self-assembled InGaAs quantum dot at liquid helium temperature. (b) Saturation of the absorption as a function of resonant laser power. The fit is a saturation curve of the form given in eqn 11.78 with a saturation power of 18 nW. After Kroner et al. (2008), © Elsevier, reprinted with permission.

given in Table 11.5, and illustrates the point that resonant nonlinearities are much larger than non-resonant ones. However, the price that has to be paid for the large $|n_2|$ value is the high absorption at the operating wavelength. This restricts the length of the devices that can be used, and therefore puts a limit on the maximum nonlinear phase shift that can be obtained.

The saturable absorption of quantum wells has found an important application in ultrafast laser physics. Ultrafast lasers work by the mode locking principle, and one of the methods used to achieve mode locking is to incorporate a saturable absorber into the laser cavity. By using molecular beam epitaxy, it is possible to make a semiconductor saturable absorber mirror (SESAM), which consists of semiconductor quantum well layers grown on top of a dielectric mirror. Such mirrors are now routinely used as the saturable absorber in a number of commercial mode-locked solid-state lasers.

In recent years there has been a growing interest in investigating the nonlinear properties of semiconductor quantum dots. The optical properties of quantum dots were discussed in Section 6.8, with Section 6.8.3 focusing on self-assembled III–V dots. Figure 11.12(a) shows the excitonic absorption line of a single self-assembled InGaAs quantum dot at liquid helium temperatures. A very sharp line with a linewidth of only 3.5 μeV is observed at low excitation powers. Figure 11.12(b) shows the absorption strength (i.e. absorbance) measured when the dot is irradiated with a laser tuned to resonance with the exciton. The absorbance is observed to saturate strongly as a function of increasing laser power P . The fit is a saturation curve of the form (cf. eqn 11.37):

$$\alpha(P) = \frac{\alpha_0}{1 + P/P_s}, \quad (11.78)$$

with a saturation power P_s of 18 nW. The good fit to the data shows that the quantum dot approximates well to a saturable two-level atom system, which is the starting point for the analysis of nonlinear optics in atomic physics. This is one of the reasons why quantum dots are often described as ‘solid-state atoms’, and have interesting possibilities for applications in quantum optics.

Chapter summary

- Nonlinear optical effects become important at the high power levels available from lasers. Nonlinear effects cause the optical susceptibility, and all the properties that follow from it, to depend on the magnitude of the electric field of the light.

- Nonlinear effects are characterized by the nonlinear susceptibility tensor. All the components of the second-order nonlinear susceptibility tensor are zero if the medium possesses inversion symmetry. Some crystals lack inversion symmetry and have non-zero second-order nonlinear susceptibilities. The form of the tensor is determined by the symmetry class of the crystal.
- Non-resonant nonlinearities are caused by the anharmonicity of the restoring force on the bound electrons.
- Resonant nonlinearities are caused by stimulated emission from the upper level when its population becomes significant. The saturation of interband absorption can be understood in terms of the blocking of the transitions due to the Pauli exclusion principle.
- The nonlinear polarization in a second-order effect is proportional to the product of two electric fields. This gives rise to a series of second-order effects such as frequency doubling, sum frequency generation, and difference frequency generation.
- Strong second-order nonlinear signals are only achieved when the phase-matching condition is satisfied.
- The anisotropy induced by a DC electric field in an electro-optic crystal is called the Pockels effect, and can be used to make amplitude or phase modulators.
- The nonlinear polarization in a third-order effect is proportional to the product of three fields. Third-order effects are generally known as four-wave mixing.
- The third-order susceptibility is the lowest-order non-zero term in the nonlinear response of isotropic media such as gases, liquids, and glasses.
- Third-order effects give rise to nonlinear refraction, frequency tripling, and stimulated Raman scattering. Self-phase modulation describes the nonlinear phase shift induced by a beam through the nonlinear refractive index.
- The saturation of interband transitions and excitonic absorption in semiconductors gives rise to large third-order nonlinearities.

Further reading

More extensive introductory reading on nonlinear optics may be found in Yariv (1997) or Tang (1995). Butcher and Cotter (1990) give a comprehensive introduction to the theory of nonlinear optics.

A classic discussion of the effects of the crystal symmetry on the physical properties of materials is given by

Nye (1957).

The optical nonlinearities due to excitons are reviewed in Chemla (1985). A more detailed review on excitonic nonlinearities in quantum wells is given by Schmitt-Rink et al. (1989).

Exercises

- (11.1) Use the Bohr model of the atom to show that the electric field experienced by an electron bound in the n th shell of a hydrogenic atom of atomic number Z is equal to $(Z^3/n^4)e/4\pi\epsilon_0 a_H^2$, where a_H is the Bohr radius of hydrogen. Make a rough estimate of this value for the valence electrons of silicon.
- (11.2) Estimate the magnitude of the electric field of the light wave for: (a) a 10 ns long pulse of energy 1 J from a Nd:glass laser with a beam diameter 5 mm; (b) a 1 mW continuous wave semiconductor laser focused into an optical fibre with a core area of $20 \mu\text{m}^2$ and a refractive index of 1.45.
- (11.3) A gas of atoms is subjected to a strong DC electric field. State with reasons whether you would expect to be able to observe any frequency doubling or not from the gas.
- (11.4) Which of the following materials would you expect to have non-zero components in the second-order nonlinear susceptibility: (a) NaCl, (b) GaAs, (c) water, (d) glass, (e) crystalline quartz, (f) ZnS (wurtzite).
- (11.5)* An intense laser is propagating through an absorbing medium as described in Section 11.2.2. The medium contains atoms with two non-degenerate levels 1 and 2, and the separation of the levels is resonant with the laser frequency. The atoms are initially in a state with $N_1 = N_0$ and $N_2 = 0$, and the laser beam is turned on at time $t = 0$.
- (a) Explain why it is not possible to get population inversion between the two levels, no matter how intense the laser is.
- (b) Show that if spontaneous emission is neglected, the time dependence of the population difference $\Delta N = N_1 - N_2$ between the two levels is given by:

$$\Delta N(t) = N_0 \exp(-2B_{12}u_\nu g(\nu)t),$$

where u_ν is the energy density of the beam, $g(\nu)$ is the spectral lineshape function, and B_{12} is the Einstein B coefficient for absorption. What does this relationship imply about the absorption coefficient?

- (11.6) The nonlinear optical coefficient tensor of a crystal with orthorhombic symmetry is given by

with orthorhombic symmetry is given by

$$d = \begin{pmatrix} 0 & 0 & 0 & d_{14} & 0 & 0 \\ 0 & 0 & 0 & 0 & d_{25} & 0 \\ 0 & 0 & 0 & 0 & 0 & d_{36} \end{pmatrix}.$$

A laser beam is incident on the crystal. The beam travels in the x - y plane, and is polarized with its electric field in the x - y plane. Show that a second harmonic beam is produced which is polarized along the z axis. Show also that the magnitude of the second harmonic beam is a maximum if the incoming beam travels at 45° to the x axis.

- (11.7) Calculate the phase-matching angle for KDP at the wavelength of the ruby laser (694 nm). The relevant refractive indices are: $n_o(694 \text{ nm}) = 1.506$, $n_o(347 \text{ nm}) = 1.534$, $n_e(347 \text{ nm}) = 1.490$.
- (11.8) Trigonal crystals with 3m symmetry (e.g. LiNbO_3) are uniaxial, and have electro-optic coefficient tensors of the form:

$$r = \begin{pmatrix} 0 & -r_{22} & r_{13} \\ 0 & r_{22} & r_{13} \\ 0 & 0 & r_{13} \\ 0 & r_{51} & 0 \\ r_{51} & 0 & 0 \\ -r_{22} & 0 & 0 \end{pmatrix},$$

when the optic axis is defined to coincide with the z axis.

- (a) Deduce the changes in the ordinary and extraordinary refractive indices (n_o and n_e respectively) when a DC electric field of magnitude $\mathcal{E}$ is applied along the optic axis.
- (b) Find the phase change induced by the electric field for a crystal of length L when light of vacuum wavelength λ is propagating along the y axis with linear polarization in the z direction.
- (c) Explain how such a device can be used as a phase modulator.
- (11.9) The anisotropy induced by an electric field applied along the z axis for crystals with the cubic zinc-blende structure ($\bar{4}3m$ symmetry) can be written in the following form:

$$\begin{aligned} n_{x'} &= n_0 + \frac{1}{2}n_0^3 r_{41} \mathcal{E}_z, \\ n_{y'} &= n_0 - \frac{1}{2}n_0^3 r_{41} \mathcal{E}_z, \\ n_z &= n_0, \end{aligned}$$

*Exercises marked with an asterisk are more challenging.

where, as with the case of the tetragonal crystal considered in Fig. 11.6(a), x' and y' are axes at 45° to the crystallographic axes x and y . n_0 is the refractive index at $\mathcal{E}_z = 0$, and r_{41} is the electro-optic coefficient.

(a) Show that when light of vacuum wavelength λ propagates through the crystal in the z direction, the phase difference introduced between the x' and y' polarizations is equal to

$$\Delta\Phi(\mathcal{E}) = \frac{2\pi}{\lambda} r_{41} n_0^3 V,$$

where V is the voltage applied across the crystal to produce the field along the z direction.

(b) Evaluate the voltage at which the phase change is equal to π for CdTe at $10.6\text{ }\mu\text{m}$, where $r_{41} = 6.8\text{ }\mu\text{m/V}$ and $n_0 = 2.6$.

(11.10) The electro-optic coefficient r_{63} of KDP at 633 nm is equal to 11 pm/V , where $n_o = 1.5074$. Calculate the voltage that must be applied to change the transmission of the modulator shown in Fig. 11.6(b) to 50% at this wavelength.

(11.11) Explain why the imaginary part of the third-order nonlinear susceptibility must be non-zero in a saturable absorber medium.

(11.12) Show that the third-order nonlinear polarization produced by a linearly polarized laser beam in an isotropic medium is always parallel to the $\mathcal{E}$ vector of the laser.

(11.13) A laser beam of wavelength $1.55\text{ }\mu\text{m}$ propagates down an optical fibre of length 10 m with a core diameter of $5\text{ }\mu\text{m}$. Calculate the power that must be launched into the fibre to produce a nonlinear phase shift of π , if the fibre has a nonlinear refractive index of $2 \times 10^{-20}\text{ m}^2\text{ W}^{-1}$.

(11.14)* A short laser pulse excites 10^{24} m^{-3} electron-hole pairs into a sample of GaAs at 4 K . Calculate the shift in the absorption edge at the band gap due to the presence of the photoexcited carriers. The band parameters for GaAs are given in Table D.2. Ignore excitonic effects.

(11.15)* The $n = 1$ heavy-hole exciton of a GaAs multiple quantum well sample has a peak absorption coefficient of $8 \times 10^5\text{ m}^{-1}$ at 847 nm . By assuming that the exciton behaves like a classical dipole oscillator, estimate the change of the refractive index that can be obtained from this sample by completely saturating the absorption. The non-resonant refractive index is 3.5 .

(11.16) (a) Use eqn 4.8 to calculate the saturation density for excitons in InP. (The relevant band structure parameters for InP are given in Table D.2, and the dielectric constant is 12.5 .)

(b) If the carrier recombination time is 1 ns , and the absorption coefficient is 10^6 m^{-1} , estimate the saturation intensity when a laser is tuned to the exciton wavelength.